LINUX COMMAND TERMINAL

CHAPTER 1 — System & User Identification

whoami

Definition / Use: Prints the effective username of the currently logged-in user.

Syntax: whoami

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ whoami updi_khadir
```

uname -r

Definition / Use: Prints the current Linux kernel release version.

Syntax: uname -r

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uname -r 6.8.0-generic
```

uname -s

Definition / Use: Prints the kernel name of the operating system.

Syntax: uname -s

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uname -s Linux
```

uname -m

Definition / Use: Prints the machine hardware architecture name.

Syntax: uname -m

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uname -m x86_64
```

uname -n

Definition / Use: Prints the network node hostname of the system.

Syntax: uname -n

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uname -n updi_khadir-VMware-Virtual-Platform
```

uname -a

Definition / Use: Prints all system and kernel information in a single line.

Syntax: uname -a

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uname -a Linux updi_khadir-VMware-Virtual-Platform 6.8.0-generic x86_64 GNU/Linux
```

CHAPTER 2 — 1. Basic Commands

pwd

Definition / Use: Prints the absolute path of the current working directory.

Syntax: pwd

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ pwd /home/updi_khadir
```

ls

Definition / Use: Lists files and directories in the current directory.

Syntax: ls

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls Documents Downloads notes.txt todo.md
```

ls /

Definition / Use: Lists files and directories in the root directory.

Syntax: ls /

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls / bin boot dev etc home lib media mnt opt proc root run sbin sys tmp usr var
```

ls -a

Definition / Use: Lists all files and directories in the current directory, including hidden ones.

Syntax: ls -a

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls -a . .. .bash_history .bashrc .cache .config Documents Downloads notes.txt todo.md
```

ls -l

Definition / Use: Lists files and directories in long format showing permissions, owner, size, and date.

Syntax: ls -l

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls -l -rw-r--r-- 1 updi_khadir updi_khadir120 Nov 12 notes.txt drwxr-xr-x 2 updi_khadirupdi_khadir4096 Nov 12 Documents
```

ls -la

Definition / Use: Lists all files including hidden files in long format.

Syntax: ls -la

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls -la drwxr-xr-x 5 updi_khadir updi_khadir4096 Nov 12 . drwxr-xr-x 3 root root 4096 Nov 11 .. -rw-r--r-- 1 updi_khadirupdi_khadir120 Nov 12 notes.txt
```

ls -lh

Definition / Use: Lists files in long format with human-readable file sizes.

Syntax: ls -lh

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls -lh -rw-r--r-- 1 updi_khadir updi_khadir1.2K Nov 12 notes.txt drwxr-xr-x 2 updi_khadirupdi_khadir4.0K Nov 12 Documents
```

cd

Definition / Use: Changes the current working directory; with no argument, it returns to the home directory.

Syntax: cd

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cd updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

cd /etc

Definition / Use: Changes the current working directory to the system configuration directory /etc.

Syntax: cd /etc

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cd /etc updi_khadir@updi_khadir-VMware-Virtual-Platform:/etc$
```

cd /var/log

Definition / Use: Changes the current working directory to the system log directory /var/log.

Syntax: cd /var/log

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cd /var/log
updi_khadir@updi_khadir-VMware-Virtual-Platform:/var/log$
```

cd ~/Documents

Definition / Use: Changes the current working directory to the Documents folder inside the home directory.

Syntax: cd ~/Documents

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cd ~/Documents
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/Documents$
```

cd ../Downloads

Definition / Use: Moves up one directory level and enters the Downloads directory.

Syntax: cd ../Downloads

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/Documents$ cd ../Downloads
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/Downloads$
```

2. Help Commands

man ls

Definition / Use: Opens the manual page for the ls command.

Syntax: man ls

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ man ls LS(1) User Commands
LS(1) NAME ls - list directory contents
```

man grep

Definition / Use: Opens the manual page for the grep command.

Syntax: man grep

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ man grep GREP(1) User
Commands GREP(1) NAME grep - print lines that match patterns
```

man cd

Definition / Use: Checks the manual page for cd (returns no entry since cd is a shell built-in).

Syntax: man cd

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ man cd No manual entry for cd
```

man 1 passwd

Definition / Use: Opens Section 1 of the manual page covering the passwd user command.

Syntax: man 1 passwd

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ man 1 passwd PASSWD(1) User Commands PASSWD(1) NAME passwd - change user password
```

man 5 passwd

Definition / Use: Opens Section 5 of the manual page describing the /etc/passwd file format.

Syntax: man 5 passwd

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ man 5 passwd PASSWD(5) File Formats and Conversions PASSWD(5) NAME passwd - password file
```

mkdir --help

Definition / Use: Displays quick usage help and options for the mkdir command.

Syntax: mkdir --help

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ mkdir --help Usage: mkdir [OPTION]... DIRECTORY... Create the DIRECTORY(ies), if they do not already exist.
```

info ls

Definition / Use: Opens the GNU Info documentation page for the ls command.

Syntax: info ls

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ info ls File: coreutils.info,
Node: ls invocation, Next: dir invocation
```

help cd

Definition / Use: Displays help information for the shell built-in cd command.

Syntax: help cd

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ help cd cd: cd [-L|[-P [-e]]
[-@]] [dir] Change the shell working directory.
```

3. History and Terminal Commands

history

Definition / Use: Displays commands previously entered in the shell session.

Syntax: history

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ history  1 pwd 2 ls 3 echo
$USER
```

echo $SHELL

Definition / Use: Prints the path of the current default login shell.

Syntax: echo $SHELL

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo $SHELL /bin/bash
```

4. File and Directory Commands

touch notes.txt

Definition / Use: Creates an empty file named notes.txt or updates its modification timestamp.

Syntax: touch notes.txt

Practice output:

```
        updi_khadir@updi_khadir-VMware-Virtual-Platform:~$      touch notes.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

touch todo.md

Definition / Use: Creates an empty Markdown file named todo.md or updates its timestamp.

Syntax: touch todo.md

Practice output:

```
        updi_khadir@updi_khadir-VMware-Virtual-Platform:~$      touch todo.md
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

touch new file

Definition / Use: Creates two separate empty files named new and file because the space is unquoted.

Syntax: touch new file

Practice output:

```
        updi_khadir@updi_khadir-VMware-Virtual-Platform:~$      touch new file
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

touch "new file"

Definition / Use: Creates a single file whose filename contains a space using quotes.

Syntax: touch "new file"

Practice output:

```
        updi_khadir@updi_khadir-VMware-Virtual-Platform:~$      touch "new file"
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

mkdir week3

Definition / Use: Creates a new directory named week3 in the current location.

Syntax: mkdir week3

Practice output:

```
        updi_khadir@updi_khadir-VMware-Virtual-Platform:~$      mkdir week3
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

mkdir -p projects/linux

Definition / Use: Creates nested directories including parent directories if they do not exist.

Syntax: mkdir -p projects/linux

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ mkdir -p projects/linux
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

rmdir week3

Definition / Use: Removes an empty directory named week3.

Syntax: rmdir week3

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ rmdir week3
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

rmdir projects

Definition / Use: Attempts to remove a directory; fails if the directory is not empty.

Syntax: rmdir projects

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ rmdir projects rmdir: failed
to remove 'projects': Directory not empty
```

cp notes.txt notes.bak

Definition / Use: Copies the contents of notes.txt into a new backup file notes.bak.

Syntax: cp notes.txt notes.bak

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cp notes.txt notes.bak
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

mv notes.txt old.txt

Definition / Use: Renames or moves notes.txt to old.txt.

Syntax: mv notes.txt old.txt

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ mv notes.txt old.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

rm notes.bak

Definition / Use: Permanently removes the specified file notes.bak.

Syntax: rm notes.bak

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ rm notes.bak
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

rm -i notes.bak

Definition / Use: Prompts for interactive confirmation before removing the file.

Syntax: rm -i notes.bak

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ rm -i notes.bak rm: remove regular file 'notes.bak'? y
```

5. Variables and PATH Commands

COURSE=linux

Definition / Use: Assigns the value linux to the local shell variable COURSE.

Syntax: COURSE=linux

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ COURSE=linux
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

echo $COURSE

Definition / Use: Prints the value stored in the shell variable COURSE.

Syntax: echo $COURSE

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo $COURSE linux
```

echo $HOME $USER

Definition / Use: Prints the current user's home directory path and username environment variables.

Syntax: echo $HOME $USER

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo $HOME $USER
/home/updi_khadir updi_khadir
```

export COURSE

Definition / Use: Exports a shell variable so child processes inherit it.

Syntax: export COURSE

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ export COURSE
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

echo $PATH

Definition / Use: Displays the colon-separated list of directories searched for executable commands.

Syntax: echo $PATH

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
```

ls /usr/bin/ls

Definition / Use: Verifies the executable binary path of the ls command.

Syntax: ls /usr/bin/ls

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls /usr/bin/ls /usr/bin/ls
```

sudo apt install sl

Definition / Use: Installs the sl package using elevated administrative privileges.

Syntax: sudo apt install sl

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sudo apt install sl Reading package lists... Done Building dependency tree... Done
```

6. Quoting and Escaping Commands

echo Hello $USER

Definition / Use: Expands the $USER variable and prints the greeting.

Syntax: echo Hello $USER

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo Hello $USER Hello updi_khadir
```

echo "Hello $USER"

Definition / Use: Double quotes preserve spaces while allowing variable expansion.

Syntax: echo "Hello $USER"

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo "Hello $USER" Hello updi_khadir
```

echo 'Hello $USER'

Definition / Use: Single quotes prevent variable expansion and treat all characters literally.

Syntax: echo 'Hello $USER'

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo 'Hello $USER' Hello $USER
```

cd lab\ report

Definition / Use: Escapes the space character using a backslash to enter a directory with spaces in its name.

Syntax: cd lab\ report

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cd lab\ report
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/lab report$
```

echo \$HOME

Definition / Use: Escapes the dollar sign so $HOME is printed literally instead of expanding.

Syntax: echo \$HOME

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo \$HOME $HOME
```

touch notes*.txt

Definition / Use: Escapes the asterisk wildcard to create a file literally named notes*.txt.

Syntax: touch notes*.txt

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ touch notes\*.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

7. Wildcard Commands

ls *.txt

Definition / Use: Lists all files ending with the .txt extension using the * wildcard.

Syntax: ls *.txt

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls *.txt lab1.txt notes.txt
```

ls lab?.txt

Definition / Use: Matches filenames starting with lab followed by any single character and .txt.

Syntax: ls lab?.txt

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls lab?.txt lab1.txt lab2.txt
```

ls lab[12].txt

Definition / Use: Matches filenames containing either 1 or 2 in the specified character position.

Syntax: ls lab[12].txt

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls lab[12].txt lab1.txt
lab2.txt
```

ls *[0-9].jpg

Definition / Use: Lists all .jpg files whose names end with a numeric digit from 0 to 9.

Syntax: ls *[0-9].jpg

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls *[0-9].jpg photo1.jpg
screen2.jpg
```

mv *.jpg Pictures/

Definition / Use: Moves all files ending in .jpg into the Pictures directory.

Syntax: mv *.jpg Pictures/

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ mv *.jpg Pictures/
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

rm *

Definition / Use: Removes all non-hidden files in the current directory.

Syntax: rm *

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/temp$ rm *
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/temp$
```

8. Short and Long Options & Other Commands

ls --all

Definition / Use: Long-option equivalent of ls -a to display all files including hidden entries.

Syntax: ls --all

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls --all . .. .bashrc
.profile Documents Downloads notes.txt
```

ls --human-readable -l

Definition / Use: Displays file sizes in human-readable format using the long option flag.

Syntax: ls --human-readable -l

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls --human-readable -l -rw-r-
-r-- 1 updi_khadirupdi_khadir1.2K Nov 12 notes.txt
```

ls --all --format=long

Definition / Use: Combines multiple long options to list all files in detailed long format.

Syntax: ls --all --format=long

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls --all --format=long drwxr-
xr-x 5 updi_khadirupdi_khadir4.0K Nov 12 . -rw-r--r-- 1 updi_khadirupdi_khadir120 Nov
12 notes.txt
```

cp /etc/hostname .

Definition / Use: Copies the system hostname file into the current working directory.

Syntax: cp /etc/hostname .

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cp /etc/hostname .
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

cd ../..

Definition / Use: Moves up two directory levels in the filesystem hierarchy.

Syntax: cd ../..

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/projects/linux$ cd ../..
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

source ~/.bashrc

Definition / Use: Reloads and executes commands from the ~/.bashrc configuration file in the current shell.

Syntax: source ~/.bashrc

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ source ~/.bashrc
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

CHAPTER 3 — 3.1 Viewing and Searching Files

cat notes.txt

Definition / Use: Displays or combines file contents directly in the terminal.

Syntax: cat [OPTIONS] [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cat notes.txt Linux practice
notes for week 3.
```

cat -n notes.txt

Definition / Use: Displays file contents with line numbers prefixed to every line.

Syntax: cat -n [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cat -n notes.txt  1 Linux
practice notes for week 3. 2 Second line of notes.
```

less notes.txt

Definition / Use: Views text one screen at a time with forward and backward navigation.

Syntax: less [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ less notes.txt Linux practice notes for week 3. (END)
```

more notes.txt

Definition / Use: Paginates text output one screenful at a time.

Syntax: more [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ more notes.txt Linux practice notes for week 3.
```

head -n 5 notes.txt

Definition / Use: Displays the first 5 lines of a file or input stream.

Syntax: head -n [COUNT] [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ head -n 5 notes.txt Line 1: Introduction to Linux Line 2: Basic file navigation
```

tail -n 5 notes.txt

Definition / Use: Displays the last 5 lines of a file or input stream.

Syntax: tail -n [COUNT] [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ tail -n 5 notes.txt Line 9: Permissions overview Line 10: End of lab summary
```

tail -f /var/log/syslog

Definition / Use: Follows newly appended lines in a log file in real time.

Syntax: tail -f [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ tail -f /var/log/syslog Nov 12 04:10:01 updi_khadir-VMware CRON[2104]: (root) CMD (run-parts)
```

file notes.txt

Definition / Use: Identifies the file type and encoding of notes.txt.

Syntax: file [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ file notes.txt notes.txt: ASCII text
```

stat notes.txt

Definition / Use: Displays detailed metadata including inode, permissions, size, and timestamps.

Syntax: stat [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ stat notes.txt  File: notes.txt Size: 120 Blocks: 8 IO Block: 4096 regular file Access: (0644/-rw-r--r--) Uid: ( 1000/updi_khadir) Gid: ( 1000/updi_khadir)
```

which ls

Definition / Use: Shows the full path of an executable command located via PATH.

Syntax: which [COMMAND]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ which ls /usr/bin/ls
```

type cd

Definition / Use: Identifies whether a command is a shell built-in, alias, function, or binary.

Syntax: type [COMMAND]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ type cd cd is a shell builtin
```

whereis bash

Definition / Use: Locates the binary, source, and manual page files for a command.

Syntax: whereis [COMMAND]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ whereis bash bash:
/usr/bin/bash /etc/bash.bashrc /usr/share/man/man1/bash.1.gz
```

find . -name "*.txt"

Definition / Use: Searches directory trees recursively for files matching a case-sensitive name pattern.

Syntax: find [PATH] -name [PATTERN]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ find . -name "*.txt"
./notes.txt ./Documents/lab.txt
```

find . -iname "notes*"

Definition / Use: Searches directory trees using a case-insensitive filename pattern.

Syntax: find [PATH] -iname [PATTERN]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ find . -iname "notes*"
./notes.txt ./Notes_Backup.txt
```

find . -type f

Definition / Use: Finds only regular files (-type f) or directories (-type d) within a path.

Syntax: find [PATH] -type [f|d]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ find . -type f ./notes.txt
./Documents/lab.txt
```

find . -size +1M

Definition / Use: Finds files larger than 1 megabyte or modified within specific minutes (-mmin).

Syntax: find [PATH] -size +1M

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ find . -size +1M
./Downloads/ubuntu-guide.pdf
```

locate notes.txt

Definition / Use: Quickly finds file paths using a prebuilt filename database updated via updatedb.

Syntax: locate [PATTERN]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ locate notes.txt
/home/updi_khadir/notes.txt
```

grep -n "Linux" notes.txt

Definition / Use: Searches text for lines matching a pattern and prints matching line numbers (-i ignores case, -v inverts match, -r searches recursively).

Syntax: grep -n [PATTERN] [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ grep -n "Linux" notes.txt
1:Line 1: Introduction to Linux
```

grep -ri "error" /var/log/

Definition / Use: Recursively searches a directory case-insensitively for a text pattern.

Syntax: grep -ri [PATTERN] [DIRECTORY]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ grep -ri "error"
/var/log/syslog Nov 12 03:45:12 updi_khadir-VMware kernel: [ 1.204] ACPI Error:
AE_NOT_FOUND
```

3.2 Redirection, Pipes and Text Processing

echo "Hello" > out.txt

Definition / Use: Redirects standard output to a file, overwriting existing contents (>> appends, 2> redirects errors, < reads standard input).

Syntax: [COMMAND] > [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo "Hello" > out.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

echo "World" >> out.txt

Definition / Use: Appends standard output to the end of an existing file without overwriting it.

Syntax: [COMMAND] >> [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo "World" >> out.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

ls /nofile 2> err.log

Definition / Use: Redirects standard error messages into a separate log file.

Syntax: [COMMAND] 2> [ERROR_FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls /nofile 2> err.log
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

ls -l | grep "txt"

Definition / Use: Uses the pipe operator (|) to pass the output of one command as input to another.

Syntax: [CMD1] | [CMD2]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls -l | grep "txt" -rw-r--r--
1 updi_khadirupdi_khadir120 Nov 12 notes.txt
```

mkdir test && cd test

Definition / Use: Command chaining operators: ; runs sequentially, && runs second only if first succeeds, || runs second only if first fails.

Syntax: [CMD1] && [CMD2]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ mkdir test && cd test
updi_khadir@updi_khadir-VMware-Virtual-Platform:~/test$
```

echo $?

Definition / Use: Prints the exit status code of the most recently executed command (0 indicates success).

Syntax: echo $?

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo $? 0
```

sort -rn numbers.txt

Definition / Use: Sorts lines of text numerically (-n) and in reverse descending order (-r).

Syntax: sort -rn [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sort -rn numbers.txt 100 42 7
```

uniq -c names.txt

Definition / Use: Filters adjacent duplicate lines and prefixes each line with its occurrence count (-c) or shows duplicates (-d).

Syntax: uniq -c [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uniq -c names.txt  2 alpha 1
bravo
```

cut -d: -f1 /etc/passwd

Definition / Use: Extracts specific fields (-f) separated by a delimiter (-d) or character positions (-c) from each line.

Syntax: cut -d[DELIM] -f[FIELDS] [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cut -d: -f1 /etc/passwd |
head -n 3 root daemon bin
```

paste file1.txt file2.txt

Definition / Use: Merges corresponding lines from multiple files side by side (-d sets delimiter, -s serializes).

Syntax: paste [FILE1] [FILE2]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ paste file1.txt file2.txt
alpha 10 bravo 20
```

wc -l notes.txt

Definition / Use: Counts lines (-l), words (-w), bytes (-c), or characters (-m) in a file.

Syntax: wc -l [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ wc -l notes.txt 10 notes.txt
```

tr 'a-z' 'A-Z' < notes.txt

Definition / Use: Translates, deletes (-d), or squeezes (-s) characters from standard input.

Syntax: tr [SET1] [SET2]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo "linux" | tr 'a-z' 'A-Z'
LINUX
```

3.3 Archives and Compression

tar -cvf archive.tar Documents/

Definition / Use: Creates (-c) a tar archive file (-f) with verbose output (-v); use -tf to list and -xf to extract.

Syntax: tar -cvf [ARCHIVE.tar] [FILES]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ tar -cvf archive.tar
Documents/ Documents/ Documents/lab.txt
```

tar -czf backup.tar.gz Documents/

Definition / Use: Creates a compressed tar archive using gzip (-z), bzip2 (-j), or xz (-J).

Syntax: tar -czf [ARCHIVE.tar.gz] [FILES]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ tar -czf backup.tar.gz
Documents/ updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

gzip notes.txt && gunzip notes.txt.gz

Definition / Use: Compresses files into .gz format using gzip and decompresses them using gunzip (-c writes to stdout).

Syntax: gzip [FILE] / gunzip [FILE.gz]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ gzip notes.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

bzip2 notes.txt && bunzip2 notes.txt.bz2

Definition / Use: Compresses and decompresses files using the bzip2 algorithm (bzcat views compressed text).

Syntax: bzip2 [FILE] / bunzip2 [FILE.bz2]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ bzip2 notes.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

xz notes.txt && unxz notes.txt.xz

Definition / Use: Compresses and decompresses files using the high-ratio xz format (xzcat views contents).

Syntax: xz [FILE] / unxz [FILE.xz]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ xz notes.txt
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

zip -r docs.zip Documents/

Definition / Use: Creates a ZIP archive recursively (-r); use unzip -l to list contents and unzip -d to extract to a directory.

Syntax: zip -r [ARCHIVE.zip] [DIRECTORY]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ zip -r docs.zip Documents/
adding: Documents/ (stored 0%) adding: Documents/lab.txt (deflated 12%)
```

3.4 Text Editors & Shortcuts

nano notes.txt

Definition / Use: Opens the beginner-friendly Nano terminal text editor. Key shortcuts: Ctrl+O (Save), Ctrl+X (Exit), Ctrl+W (Search), Ctrl+K (Cut), Ctrl+U (Paste), Alt+U (Undo).

Syntax: nano [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ nano notes.txt  GNU nano 7.2
notes.txt Linux practice notes for week 3.
```

vim notes.txt

Definition / Use: Opens the Vim modal text editor (also vi and vimtutor). Key commands: i (Insert mode), Esc (Normal mode), :w (Save), :q (Quit), :wq (Save & Quit), :q! (Quit without saving).

Syntax: vim [FILE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ vim notes.txt Linux practice
notes for week 3. ~ :wq
```

3.5 Hard Links and Symbolic Links

ln -s notes.txt shortcut.txt

Definition / Use: Creates a symbolic link (-s) pointing to a target file, or a hard link without -s; verify with ls -li.

Syntax: ln -s [TARGET] [LINK_NAME]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ln -s notes.txt shortcut.txt
&& ls -l shortcut.txt lrwxrwxrwx 1 updi_khadirupdi_khadir9 Nov 12 shortcut.txt ->
notes.txt
```

CHAPTER 4 — 4.1 Processes and Jobs

echo $$

Definition / Use: Prints the Process ID (PID) of the current shell session.

Syntax: echo $$

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ echo $$ 1842
```

ps aux

Definition / Use: Displays detailed information about all running processes across all users (also ps -ef and ps -o pid,ppid,comm).

Syntax: ps aux

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ps -o pid,ppid,comm  PID PPID
COMMAND 1842 1800 bash 1901 1842 ps
```

ps aux --sort=-%mem | head -n 5

Definition / Use: Lists the top memory-consuming processes sorted in descending order.

Syntax: ps aux --sort=-%mem | head -n 5

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ps -p 1 -o pid,user,comm  PID
USER COMMAND 1 root systemd
```

pstree -p $$

Definition / Use: Displays running processes as a parent-child hierarchy tree with process IDs.

Syntax: pstree -p $$

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ pstree -p $$
bash(1842)──pstree(1945)
```

top / htop

Definition / Use: Opens an interactive real-time monitor of CPU, memory, and running processes (use -u username to filter by user).

Syntax: top -u `updi_khadi`

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ command -v htop /usr/bin/htop
```

sleep 30 &

Definition / Use: Runs a process in the background using &; manage jobs with jobs -l, bg %1, and fg %1.

Syntax: sleep 30 &

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sleep 30 & [1] 1962
```

jobs -l

Definition / Use: Lists active background and stopped jobs started in the current shell along with their PIDs.

Syntax: jobs -l

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ jobs -l [1]+ 1962 Running
sleep 30 &
```

4.2 Signals, Kill and Priority

kill -l

Definition / Use: Lists available process signals such as HUP (1), INT (2), KILL (9), and TERM (15).

Syntax: kill -l

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ kill -l 15 TERM
```

pgrep -a sleep

Definition / Use: Searches for running processes by name and displays their PIDs and full command lines.

Syntax: pgrep -a [PROCESS_NAME]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ pgrep -a sleep 1962 sleep 30
```

kill -TERM 1962

Definition / Use: Sends a termination signal to a process ID (use -9 for force kill, or pkill/killall by process name).

Syntax: kill -TERM [PID]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ kill -TERM 1962 [1]+
Terminated sleep 30
```

nice -n 10 sleep 300 &

Definition / Use: Starts a new process with a modified scheduling niceness priority value.

Syntax: nice -n [VALUE] [COMMAND]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ nice -n 10 sleep 300 & [1]
2015
```

renice -n 15 -p 2015

Definition / Use: Alters the scheduling priority of an already running process by its PID.

Syntax: renice -n [VALUE] -p [PID]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ renice -n 15 -p 2015 2015
(process ID) old priority 10, new priority 15
```

4.3 System Information

uname -a

Definition / Use: Displays full kernel (-r), machine architecture (-m), nodename (-n), and OS details.

Syntax: uname -a

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uname -a Linux updi_khadir-VMware-Virtual-Platform 6.8.0-generic x86_64 GNU/Linux
```

hostnamectl hostname

Definition / Use: Queries or changes the system hostname and related machine metadata.

Syntax: hostnamectl hostname

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ hostnamectl hostname
updi_khadir-VMware-Virtual-Platform
```

uptime -p

Definition / Use: Shows how long the system has been running (-p for pretty format, -s for boot timestamp).

Syntax: uptime -p

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ uptime -p up 2 hours, 14 minutes
```

nproc

Definition / Use: Prints the number of processing units (CPU cores) available to the system.

Syntax: nproc

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ nproc 4
```

free -h

Definition / Use: Displays system memory and swap usage in human-readable units (-m displays in megabytes).

Syntax: free -h

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ free -h total used free shared buff/cache available Mem: 7.7Gi 2.1Gi 3.8Gi 120Mi 1.8Gi 5.2Gi Swap: 2.0Gi 0B 2.0Gi
```

lscpu

Definition / Use: Displays detailed CPU architecture, core count, and processor model information.

Syntax: lscpu

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ grep -m 1 "model name" /proc/cpuinfo model name : Intel(R) Core(TM) processor
```

lsblk -e 7 -d -o NAME,SIZE,MODEL

Definition / Use: Lists block storage devices, disks, and partitions while excluding loop devices (-e 7).

Syntax: lsblk -e 7 -d -o NAME,SIZE,MODEL

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ lsblk -e 7 -d -o NAME,SIZE,MODEL NAME SIZE MODEL sda 40G VMware Virtual S
```

cat /proc/uptime && cat /proc/1/comm

Definition / Use: Reads kernel runtime statistics from /proc/uptime, /proc/meminfo, and /proc/1/comm.

Syntax: cat /proc/1/comm

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ cat /proc/1/comm systemd
```

ls /sys/class/net

Definition / Use: Inspects network interfaces and block devices exposed through the /sys virtual filesystem.

Syntax: ls /sys/class/net

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ ls /sys/class/net ens33 lo
```

4.4 Services with systemd

systemctl status cron

Definition / Use: Displays runtime status, PID, and recent log entries for a systemd service unit.

Syntax: systemctl status [SERVICE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ systemctl status cron ● cron.service - Regular background program processing daemon Loaded: loaded (/lib/systemd/system/cron.service; enabled) Active: active (running)
```

systemctl is-active cron && systemctl is-enabled cron

Definition / Use: Checks whether a systemd service is currently active and enabled to start at boot.

Syntax: systemctl is-active [SERVICE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ systemctl is-active cron active
```

sudo systemctl restart cron

Definition / Use: Controls systemd services: start, stop, restart, reload, enable --now, or disable.

Syntax: sudo systemctl restart [SERVICE]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sudo systemctl restart cron updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

systemctl list-units --type=service --state=running

Definition / Use: Lists active systemd service units filtered by state (running or failed).

Syntax: systemctl list-units --type=service --state=running

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ systemctl show -p MainPID cron MainPID=842
```

4.5 Logs, Boot and Troubleshooting

journalctl -n 3

Definition / Use: Queries systemd journal logs (-b current boot, -p err errors only, -u unit name, -f follow live).

Syntax: journalctl -n [LINES]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ journalctl -u cron -n 2 Nov 12 04:00:01 updi_khadir-VMware systemd[1]: Started Regular background program processing daemon.
```

logger "week 8 test from student"

Definition / Use: Writes a custom message directly into the system log (/var/log/syslog).

Syntax: logger "[MESSAGE]"

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ logger "week 8 test from student" updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

last -n 2

Definition / Use: Displays a list of the most recent user login and reboot sessions.

Syntax: last -n [COUNT]

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ last -n 2 updi_khadirpts/0 :0 Thu Nov 12 02:15 still logged in reboot system boot 6.8.0-generic Thu Nov 12 02:10 still running
```

sudo dmesg -T | tail -n 2

Definition / Use: Prints kernel ring buffer messages with human-readable timestamps (-T).

Syntax: sudo dmesg -T | tail -n 2

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sudo dmesg -T | tail -n 2
[Thu Nov 12 02:10:04 2026] e1000: ens33 NIC Link is Up 1000 Mbps Full Duplex
```

sudo shutdown -r +5 "Rebooting for the lab"

Definition / Use: Schedules a system reboot (+5 minutes) or poweroff; cancel a scheduled shutdown with sudo shutdown -c.

Syntax: sudo shutdown -r +5 "[MESSAGE]"

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sudo shutdown -c
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```

sudo sshd -t

Definition / Use: Tests the SSH daemon configuration file syntax for validity before restarting the service.

Syntax: sudo sshd -t

Practice output:

```
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$ sudo sshd -t
updi_khadir@updi_khadir-VMware-Virtual-Platform:~$
```